Nonsingularity of the periodic random-walk system

A solution to MathDB 357004

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Abstract

We prove the nonsingularity conjecture of Gervé and Grigutis for every positive integer premium and every number of seasons, under the source's net profit, positive zero-mass, and simple-root assumptions. A null vector would produce generating functions continuous on the closed unit disk. Their coefficients tend to zero and satisfy a positive recurrence. An elementary maximum principle then forces the null vector to vanish. Roots on the unit circle are included.

Task 1: the original problem and its full solution

Let $X_1, \dots, X_N$ be independent, nonnegative integer-valued random variables, and let $\kappa, N \geq 1$. We retain the net profit condition used to construct system (16) in [1]:

$$\mu = \sum_{j=1}^{N} \mathbb{E}X_j < \kappa N, \qquad \mathbb{P}(X_1 + \cdots + X_N = 0) > 0. \tag{1}$$

Write $G_X(z) = \mathbb{E}z^X$. The source proves that $G_{X_1+\cdots+X_N}(z) - z^{\kappa N}$ has exactly $\kappa N - 1$ zeros other than 1 in the closed unit disk, counted with multiplicity [1, Lemma 4]. In Conjecture 1 these zeros $\alpha_1, \dots, \alpha_{\kappa N-1}$ are assumed simple. The conjecture states that the coefficient matrix of system (16) is nonsingular.

We specify that matrix without its large block display. This also fixes all cyclic indices. Set

$$Y_1 = X_N, \quad Y_j = X_{j-1} \ (2 \leq j \leq N), \qquad g_j(z) = G_{Y_j}(z), \quad p_{j,r} = \mathbb{P}(Y_j = r),$$

and $F_j(r) = \sum_{q=0}^{r} p_{j,q}$. In particular $p_{j,0} > 0$, by independence and (1). For a vector $v = (v_{j,i})_{1 \leq j \leq N,\ 0 \leq i < \kappa}$, define

$$B_j(z) = \sum_{i=0}^{\kappa-1} v_{j,i} \sum_{r=i}^{\kappa-1} z^r F_j(r-i). \tag{2}$$

The first $\kappa N - 1$ rows of the source matrix A act by

$$(Av)_\ell = \sum_{j=1}^{N} \frac{\prod_{r=1}^{j-1} g_r(\alpha_\ell)}{\alpha_\ell^{\kappa(j-1)}} B_j(\alpha_\ell), \qquad 1 \leq \ell < \kappa N. \tag{3}$$

Its final row acts by

$$(Av)_{\kappa N} = \sum_{j=1}^{N} B_j(1). \tag{4}$$

Indeed, $\sum_{r=i}^{\kappa-1} F_j(r-i) = \sum_{q=0}^{\kappa-1-i} p_{j,q}(\kappa-i-q)$, which is exactly the final row printed in system (16). Equations (3) and (4) are respectively the homogeneous forms of source equations (13) and (11).

Theorem 1. *Under* (1) *and the simple-root hypothesis above, the κN by κN matrix A is nonsingular.*

Two elementary uniqueness facts

Lemma 1. *If C is holomorphic in the open unit disk and continuous on its closure, its Taylor coefficients $h(n) = [z^n]C(z)$ tend to zero.*

Proof. Put $M = \max_{|z|\le 1} |C(z)|$. Cauchy's estimate gives $|h(n)| \le M$. Uniform continuity implies $\varepsilon_\rho = \max_{|z|\le 1} |C(z) - C(\rho z)| \to 0$ as $\rho \to 1^-$. Again by Cauchy's estimate, $|(1-\rho^n)h(n)| \le \varepsilon_\rho$, so $|h(n)| \le M\rho^n + \varepsilon_\rho$. First let $n \to \infty$, then $\rho \to 1^-$. □

Lemma 2. *Let $h_1, \dots, h_N$ be complex sequences on the nonnegative integers, each tending to zero. Extend the indices periodically, so $h_0 = h_N$. Suppose $p_{j,r} \ge 0$, $\sum_{r\ge 0} p_{j,r} = 1$, $p_{j,0} > 0$, and*

$$h_{j-1}(u) = \sum_{r=0}^{u+\kappa} p_{j,r} h_j(u+\kappa-r), \qquad u \ge 0. \tag{5}$$

Then all the sequences vanish identically.

Proof. If not, their common supremum in modulus is a positive number M. It is attained at some $(j-1, u)$, since there are finitely many sequences and each tends to zero. Taking absolute values in (5) bounds the right-hand side by M. Because $p_{j,0} > 0$, equality is possible only if $|h_j(u+\kappa)| = M$: otherwise that positive-weight term makes the bound strictly smaller than M. Repeating this argument propagates the modulus M to arguments $u + q\kappa$ for every $q \ge 0$, contradicting convergence to zero. □

From a null vector to a decaying recurrence

Proof of Theorem 1. Suppose $Av = 0$, and form (2). Put

$$K(z) = \prod_{r=1}^{N} g_r(z) - z^{\kappa N}.$$

Extend g_j, B_j periodically in j, and, for $1 \le j \le N$, set

$$H_j(z) = \sum_{\ell=1}^{N} z^{\kappa(N-\ell)} \left(\prod_{r=1}^{\ell-1} g_{j+r}(z) \right) B_{j+\ell}(z). \tag{6}$$

Direct cancellation of consecutive terms gives the algebraic identity

$$g_j H_j - z^\kappa H_{j-1} = K B_j, \qquad H_0 = H_N. \tag{7}$$

For $N = 1$ this reads $(g_1 - z^\kappa)B_1 = KB_1$, so the one-season case is included.

In particular

$$H_N(z) = \sum_{j=1}^{N} z^{\kappa(N-j)} \left(\prod_{r=1}^{j-1} g_r(z) \right) B_j(z).$$

The root equations (3) imply $H_N(\alpha_\ell) = 0$. No α_ℓ is zero, since $K(0) = \prod_j p_{j,0} > 0$. At any such root, (7) propagates the vanishing backwards from H_N to every H_j. At 1 we have

$$H_j(1) = \sum_{r=1}^{N} B_r(1) = 0$$

by (4). Thus every H_j vanishes at every zero of K in the closed unit disk.

Define $C_j = H_j/K$ away from these zeros. The interior singularities are removable by the simple-root hypothesis. We check the boundary carefully: finite mean in (1) implies that each g_j and g_j' extends continuously to the closed disk, by absolute uniform convergence of the derivative series. The same holds for H_j, K and their first derivatives. Also $K'(1) = \mu - \kappa N < 0$. More generally, if $|\alpha| = 1$ and $K(\alpha) = 0$, equality in the triangle inequality for the probability generating function of $S = \sum_j Y_j$, together with $\mathbb{P}(S = 0) > 0$, gives

$$\alpha^s = 1 \text{ whenever } \mathbb{P}(S = s) > 0, \qquad \alpha^{\kappa N} = 1.$$

Consequently

$$K'(\alpha) = \alpha^{-1}(\mu - \kappa N) \neq 0. \tag{8}$$

For any function f here vanishing at α, integration along the segment from α to z gives $f(z)/(z - \alpha) \to f'(\alpha)$ as $z \to \alpha$ within the closed disk. Applying this to H_j and K proves that C_j extends continuously at every boundary root. Hence C_j is holomorphic in the disk and continuous on its closure. Lemma 1 implies

$$h_j(n) := [z^n]C_j(z) \longrightarrow 0.$$

Dividing (7) by K and then continuing through its removable zeros gives

$$g_j(z)C_j(z) - z^\kappa C_{j-1}(z) = B_j(z). \tag{9}$$

Since $\deg B_j < \kappa$, comparison of the coefficient of $z^{u+\kappa}$ in (9) gives exactly (5). Lemma 2 forces all $h_j(n)$, hence all C_j and all B_j, to vanish. Finally the map from $(v_{j,0}, \ldots, v_{j,\kappa-1})$ to the coefficients of B_j is triangular, with diagonal $F_j(0) = p_{j,0} > 0$. Therefore every $v_{j,i} = 0$. The square matrix A has trivial kernel and is nonsingular. □

The explicitly stated three-season inequality. For $\kappa = 1, N = 3$, put $u_\ell = G_{X_3}(\alpha_\ell)/\alpha_\ell$ and $v_\ell = G_{X_3+X_1}(\alpha_\ell)/\alpha_\ell^2$. Factoring the positive zero-masses from the three columns of the source matrix (26), then subtracting its final row from the first two, gives

$$\det A = \mathbb{P}(X_1 = 0)\mathbb{P}(X_2 = 0)\mathbb{P}(X_3 = 0)\big((u_1 - 1)(v_2 - 1) - (u_2 - 1)(v_1 - 1)\big).$$

Theorem 1 makes this nonzero, which is precisely the additional inequality requested in the conjecture.

Tasks 2–5: proposed follow-up directions

These are further questions, not additional solved claims.

Task 2. Remove the simple-root restriction by determining the exact nonsingularity statement for the source's confluent matrix, whose repeated root rows are replaced by derivative rows.

Task 3. When some $\mathbb{P}(X_j = 0)$ vanish, characterize which columns and root constraints must be removed to obtain a square nonsingular system, including seasons whose minimum claim exceeds κ.

Task 4. Find quantitative bounds for the smallest singular value of the row-normalized system matrix in terms of the net profit margin, zero-masses, root separation, and a specified uniform moment bound.

Task 5. Generalize the argument to claim distributions modulated by a finite-state Markov chain, giving explicit irreducibility and boundary conditions that guarantee uniqueness of the corresponding generating-function system.

Verification and status. The accompanying exact checks cover the cyclic numerator identities, the matrix's normalization row, finite Taylor coefficient recurrences, and boundary-root identities. They supplement the analytic proof for arbitrary distributions. The published 35-page primary was read in full; Conjecture 1 remains on printed page 335. The earlier 32-page arXiv source was checked at its definition and target passages, not read in full. The dated literature audit records bounded prior-claim checks. Verification is by the authoring agent only, without independent peer review.

References

[1] S. Gervé and A. Grigutis, *Distribution of shifted discrete random walk generated by distinct random variables and applications in ruin theory*, Modern Stochastics: Theory and Applications **11**(3) (2024), 323–357. doi:10.15559/24-VMSTA249. Earlier version: arXiv:2211.14629v1 (2022).